ALEXANDRU CRISTIAN CALIN

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EDUCATION

London School of Economics Sep 2024 to Sep 2025

MSc Applicable Mathematics. Dissertation in multiagent reinforcement learning in energy markets, published on SSRN. Modules: reinforcement learning, machine learning and data mining, game theory, algorithms.

Imperial College London Jul 2024 to Aug 2024

Summer Research Student, Machine Learning and Quantitative Finance. Built a time series model in R to forecast energy commodity prices; ran stability diagnostics and traced the drivers of forecast error.

University of Manchester Sep 2020 to Jun 2024

BSc Mathematics. Dissertation in algorithmic game theory; modules in probability, statistical inference, numerical analysis and time series.

EXPERIENCE

TikTok (ByteDance) Sep 2025 to Present

Strategy and Operations Manager, TikTok Live, London

- Built a new partner pipeline across France, Spain, the UK and Romania, qualifying each prospect by revenue potential.
- Managed relationships with the leadership of 55 partner businesses, running discovery conversations and regular executive reviews to turn their goals and performance data into tailored growth plans; the best performing campaign now generates over $100k a month.
- Researched and compared four European markets to judge where growth potential was highest.
- Designed incentive experiments, measured them against a baseline and dropped three that did not change partner behaviour, refocusing effort on what worked.
- Built SQL and Python segmentation of the portfolio of 55 partner businesses to rank them by revenue potential and find the highest value segments.
- Built AI workflows with large language models that analysed partner data and drafted first versions of growth strategies, checking and correcting outputs before use.

Regnum Aegyssus Jun 2025 to Sep 2025

Financial Analyst, Romania

- Modelled volatility transmission across Eastern European equity markets in Python using rolling window multivariate estimation; the research is under journal review.

RESEARCH AND PROJECTS

Do AI Agents Play Like People? (independent project, in progress) Oct 2026 to Present

- Comparing how humans, bounded rationality models (QRE, level k, cognitive hierarchy), reinforcement learning agents and open weight language models play the same 2,416 two player games, using a public dataset of 93,460 human decisions (Zhu et al., 2025).

Multiagent Reinforcement Learning in Energy Markets (LSE MSc dissertation, SSRN) 2025

- Built a multi agent energy market environment in Python and implemented Q learning from first principles; analysed how weaker agents shape outcomes for the whole market.

Credit Default Prediction Model (LSE) May 2025 to Jun 2025

- Compared neural networks, k nearest neighbours and random forests on 30,000 credit card records, reaching an AUC-ROC of 0.82 through feature engineering and hyperparameter tuning.

Volatility Spillovers in Central European Markets (SSRN working paper)

- Estimated GARCH, BEKK and DCC models in R to measure how volatility spread between the German, Polish, Hungarian and Czech stock markets after the war in Ukraine began.

SKILLS

- **Programming and data:** Python (NumPy, pandas, scikit learn, statsmodels, matplotlib), R, SQL, Excel, Git, LaTeX.
- **Machine learning:** reinforcement learning (Q learning, policy gradients), multi agent systems, neural networks, random forests, kNN, time series forecasting; model evaluation (AUC-ROC, stability diagnostics, error analysis).
- **Quantitative:** probability theory, statistical inference, numerical analysis, time series analysis, volatility modelling (GARCH, BEKK, DCC), game theory, simulation.
- **Languages:** English (fluent), Romanian (native).